

RESEARCH INTEREST

Long's research focuses on advancing optimization techniques for multi-objective deep neural networks. He specifically investigates optimization methods to improve model robustness, safety, and helpfulness simultaneously. A key area of his work involves scaling these effectiveness measures to large-scale applications, particularly Large Language Models (LLMs).

EDUCATION

PhD	Singapore University of Technology and Design (SUTD)	09/2024 – Present
- Academic advisors: Asst. Prof. Wenxuan Zhang and Prof. Wei Lu. - Major: Artificial Intelligence.		
Bachelor	Hanoi University of Science and Technology (HUST)	08/2018 – 09/2022
- Academic advisor: Dr. Thang N. Tran - Major: Mathematics and Informatics. CPA: 3.11/4.00 (Rank 18 out of 94 in his department).		

PUBLICATIONS

- [1] **Long P. Hoang**, Wenxuan Zhang, Wei Lu, [Difficulty–Diversity Collaborative Filtering for Data-Efficient LLM Fine-Tuning](#), *In Proceedings of The International Conference on Learning Representations (ICLR)*, 2026.
- [2] Minh-Duc Nguyen, Phuong Mai Dinh, Quang-Huy Nguyen, **Long P. Hoang**, Dung D. Le, [Improving Pareto Set Learning for Expensive Multi-objective Optimization via Stein Variational Hypernetworks](#), *In Proceedings of the AAAI Conference on Artificial Intelligence*, 2025.
- [3] Tuan A. Tran[†], **Long P. Hoang**[†], Dung D. Le, Thang N. Tran, [A Framework for Controllable Pareto Front Learning with Completed Scalarization Functions and its Applications](#), *Neural Networks*, 2024.
- [4] **Long P. Hoang**, Dung D. Le, Tuan A. Tran, Thang N. Tran, [Improving Pareto Front Learning via Multi-Sample Hypernetworks](#), *In Proceedings of the AAAI Conference on Artificial Intelligence*, 2023 (Oral).

EXPERIENCE

PhD Student	The inclusive NLP Lab
Advisor: Prof. Wei Lu and Asst. Prof. Wenxuan Zhang	
<i>Difficulty-Diversity Collaborative Filtering for Data-Efficient LLM Fine-Tuning</i>	<i>04/2025 – 12/2025</i>
- Developing a collaborative filtering framework to select a small, challenging, and diverse dataset personalized for a given language model (Accepted to ICLR 2026). - Empirically validated that fine-tuning with 1,000 high-quality examples selected via our method yields better performance than using the entire 220,000-example dataset.	
Research Assistant	College of Engineering and Computer Science, VinUniversity
Advisor: Asst. Prof. Dung D. Le	

[†]Co-First Author

Expensive Multi-Objective Optimization

03/2023 – 10/2023

- Approximated the entire trade-off curve of black-box objects using Pareto Front Learning with Hypernetworks computed by Gaussian Processes (accepted to AAAI 2025).

Profiling the Pareto Front in Multi-Task Learning

02/2022 – 10/2022

- Proposed a novel method named Multi-Sample Hypernetwork to approximate the entire trade-off curve of conflicting objectives (accepted to AAAI 2023).

Undergraduate Research Assistant

School of Applied Mathematics and Informatics, HUST

Advisor: [Dr. Thang N. Tran](#)

Multi-Objective Optimization with Completed Scalarizations

02/2022 – 02/2023

- Proposed and proved the convergence of a new framework for Pareto Multi-Task Learning with Scalarization Functions in the pseudo-convex and quasiconvex assumptions (accepted to the journal Neural Networks).

Building Footprint Extraction from Remote Sensing Images

07/2021 – 10/2022

- Developed a two-stage framework, which combines U2-Net and Mask-CNN, to increase 1.8-2.5% mAP, mAR for Building Footprint Extraction, especially effective in populated areas (accepted to ICISN 2022).

ADWARDS & CERTIFICATES

- AISG PhD Fellowship, granted by AI Singapore, Sep 2024 – Sep 2028.
- 3rd Prize (Grooo International company's sponsorship) in Scientific Research Student Conference at School of Mathematics and Informatics, Hanoi University of Science and Technology, Jul 2022.
- Certificate of Completion of Developer Circles Vietnam Innovation Challenge in Data Science, sponsored by Facebook, Nov 2020.

REFERENCES

1. [Assistant Professor Wenxuan Zhang](#) (Ph.D), Information Systems Technology and Design, Singapore University of Technology and Design
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2. [Professor Wei Lu](#) (Ph.D), College of Computing and Data Science, Nanyang Technological University
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3. [Assistant Professor Dung D. Le](#) (Ph.D), College of Engineering and Computer Science, VinUniversity
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4. [Dr. Thang N. Tran](#) (Ph.D), Faculty of Applied Mathematics and Informatics, Hanoi University of Science and Technology
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